

User Manual – Football Tournament Management System

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1 Introduction and System Overview

The *Football Tournament Management System* is an application designed to organize, track, and administer football competitions. The system relies on a decoupled architecture:

- **Backend:** A REST API developed with FastAPI and connected to a relational database.
- **Frontend:** A desktop graphical interface developed in Python Tkinter, handling user interaction.
- **Orchestration:** Containerized deployment via Docker Compose ensuring infrastructure portability and stability.

Two main user roles are supported by the system:

- **Organizer Mode:** Full access to write operations (creation, modification, and deletion of tournaments, teams, and matches) via an authentication key (**X-API-Key**).
- **Viewer Mode:** Read-only mode accessible to all participants, allowing them to view schedules, scores, and real-time updated rankings.

2 Backend Deployment and Launch (Docker Compose)

The backend server and database run via Docker Compose to ensure runtime environment integrity.

Deployment Procedure

1. Open a terminal and navigate to the project root directory:

```
cd ~/football-tournament-manager
```

2. Launch the containers in the background:

```
docker compose up --build -d
```

3. Check the status of active services:

```
docker compose ps
```

3 Frontend Launch and Connection

Once the backend server is operational on port 8000, the client graphical interface can be started.

Launch Commands

```
cd frontend
source venv/bin/activate
python3 -m football_tournament
```

Startup Connection Dialog

At startup, the application presents a window requesting connection details:

- **API URL:** Specify the server address (default: <http://localhost:8000>).
- **X-API-Key:**
 - Enter the administrator key (**mykey**) to enable Organizer Mode.
 - Leave the field blank to access Viewer Mode directly.

4 Interface Navigation and Usage

The main interface is structured around a tabbed dashboard allowing seamless navigation between the system's various entities.

Tournaments Tab

- Displays the complete list of registered competitions.
- Allows viewing start dates, end dates, and the current status of each tournament.

Teams Tab

- Lists participating clubs with their unique identifier, official name, and coach/manager name.

Matches Tab

- **Global Tournament Selector:** Located at the top of the tab (active in both modes), it allows instantly filtering matches to show only those of a specific tournament or selecting "All Tournaments".
- **Score Format:** Planned matches with scores not yet defined explicitly display a dash (-). Validated scores are displayed in the **Home** - **Away** format.
- **Referee:** A dedicated column indicates the name of the referee assigned to the fixture.

Standings Tab

- Dynamically calculates team rankings per tournament based on aggregated match results (points, wins, draws, losses, goals for, goals against, and goal difference).

5 Administrative Operations and Logout Guide

Users authenticated as organizers have additional action buttons in the management tabs:

- **Scheduling a Match:** Click the + **Schedule Match** button to open the creation form, select the tournament, home and away teams, date, and save via **[SAVE MATCH]**.
- **Updating a Score:** Select a fixture in the matches table and click **Update Score** to modify the recorded goals.
- **Deletion:** Use the **Delete** button to remove an obsolete team or match from the database. All deletions strictly respect database referential integrity constraints.
- **Synchronization:** In case of external changes, use the **Refresh Data** button located in the upper toolbar to instantly refresh the local display from the API.
- **Logout:** To terminate an active session or switch user roles (e.g., from Organizer Mode back to Viewer Mode), simply close the application window or use the logout/disconnect option available in the connection panel. Reopening the app will prompt you again for credentials.